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Patent Public Search | Text View

United States Patent Application Publication

20250281682

Kind Code

A1

Publication Date

September 11, 2025

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CONTAINER FOR FLUIDS AND APPARATUS FOR TEMPERATURE CONTROL, E.G. WARMING, OF MEDICAL FLUIDS

Abstract

An extracorporeal blood circuit comprises a blood withdrawal line, a blood return line, one or more air separators, a dialysis supply line, a dialysis effluent line, one or more infusion lines, a flexible bag for enabling blood flow connected to the blood return line or to the blood withdrawal line, and a fluid conduit. The fluid conduit has a maximum width in a direction of fluid flow through the fluid conduit. The one or more deflection sections includes an entry section and an exit section. The width of the fluid conduit decreases along the direction of fluid flow through the entry section from the maximum width to a narrower width and the width of the fluid conduit increases along the direction of the fluid flow through the exit section from the narrower width to the maximum width.

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Family ID: 1000008619748

Appl. No.: 19/214945

Filed: May 21, 2025

Foreign Application Priority Data

EP	17170551.0	May. 11, 2017
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Related U.S. Application Data

parent US continuation 18380360 20231016 parent-grant-document US 12318520 child US 19214945

Publication Classification

Int. Cl.: **A61M1/16** (20060101); **A61B90/00** (20160101); **A61F7/00** (20060101); **A61M1/30** (20060101); **A61M1/36** (20060101); **F28F13/08** (20060101)

U.S. Cl.:

CPC **A61M1/1668** (20140204); **A61M1/1623** (20140204); **A61M1/166** (20140204); **A61M1/308** (20140204); **A61M1/3623** (20220501); **F28F13/08** (20130101); **A61B2090/0808** (20160201); **A61F2007/0059** (20130101); **A61M2205/0216** (20130101); **A61M2206/10** (20130101)

Background/Summary

PRIORITY CLAIM [0001] The present application is a continuation application of U.S. application Ser. No. 18/380,360, which claims priority to U.S. application Ser. No. 16/611,525, filed Nov. 7, 2019, now U.S. Pat. No. 11,786,646, which is a National Phase of International Application No. PCT/EP2018/062135, filed May 9, 2018, published as PCT Publication No. WO 2018/206718 on Nov. 15, 2018, which claims priority to EP Application No. 17170551.0, filed May 11, 2017, the entire contents of each of which are incorporated herein by reference and relied upon.

TECHNICAL FIELD

[0002] The present invention relates to a container for fluids (in particular, but not exclusively, a container for cooling or warming fluids) and to an apparatus for temperature control, e.g. warming or cooling, fluids in an apparatus for extracorporeal blood treatment.

[0003] The apparatus for extracorporeal blood treatment includes an apparatus for fluid temperature control (warming or cooling) that is configured to regulate the temperature of medical fluids, for example blood, in combination with a bag for warming or cooling medical fluids. The bag is inserted into the fluid temperature control apparatus and the temperature is regulated, for example, by heat being transferred (or removed) from a heating area of the fluids temperature control apparatus through the material of the bag and into (or from) the medical fluid flowing through the bag.

[0004] The method for controlling the extracorporeal blood treatment apparatus includes controlling the temperature of a medical fluid, for example blood, using the fluid temperature control apparatus and a bag.

BACKGROUND

[0005] In medical applications such as extracorporeal blood treatment it is often desired to regulate the temperature of medical fluids, for example in cases where the individual temperature of a medical fluid is important for particular chemical or biological reactions to take place, or for treatment to be performed under optimal conditions. Other cases include returning blood to a patient that has been withdrawn for extracorporeal treatment (e.g. dialysis, hemofiltration, hemodiafiltration).

[0006] In many cases, the flow rate of medical fluid varies over time and depends on a number of factors including treatment type, patient properties and condition, topology of the extracorporeal circuit, etc. Regulating the temperature under these varying conditions requires a robust system

facilitating reliable treatment and handling of medical fluids.

[0007] U.S. Pat. No. 6,464,666 discloses a fluid warming cassette with a stiffening frame structure and an integral handle supporting a parenteral fluid container. The fluid container is desirably thin to minimize heat exchange inefficiencies. The frame structure permits the thin fluid container to be inserted into the narrow space between fixed position warming plates of a warming unit. The frame structure has a quadrilateral shape with sides and ends. The fluid container is attached, at its periphery to the sides and ends of the frame structure, within the quadrilateral shape. Part of the frame structure is formed into a handle to assist in both the insertion and removal of the cassette from a warming unit.

[0008] U.S. 2008/0262409 discloses a system and method for manufacturing a heat exchanger. The heat exchanger comprises a casing with a serpentine pathway, a membrane enclosed by the casing, an inlet tube or valve for fluid to enter the heat exchanger, and an outlet tube or valve for fluid to exit the heat exchanger. The casing encloses the membrane, which resides between the casing and the fluid. The exterior casing possesses a wide asymmetrical serpentine pathway and particularly comprises a rigid, plastic frame. The frame consists of two halves, where the shape of the pathway is mirrored on both halves of the frame so that when the halves are sandwiched together they form a channel. The method comprises attaching a plurality of tubes or valves to a flexible container, creating an asymmetric passage in a rigid shell, enclosing the flexible container within the shell, and sealing the shell.

[0009] JP 2001120658 discloses a medical heat exchanger bag and method for manufacturing the same by which a liquid is prevented from leaking outside due to breakage caused by the internal pressure generated when the liquid flows in a liquid path, by reducing a load applied to the inside of a bent part of a medical heat exchanger bag. The bag is provided with a series of elongated channels and a series of bends that include two curved channels for fluid to flow through. Both the elongated and the curved channels have increasing or decreasing diameter along the direction of fluid flow.

[0010] Known designs bags or cassettes have shown problems with potential air retention in the internal conduit. Air retention may be caused to some extent by the presence of low speed areas in the internal conduit, for example at inlet/outlet sections, and at an outer border of the one or more bends typically included in the conduit. If the conveyed fluid is blood, then such effects entail the risk of blood clotting. Due to air retention and/or the risk of blood clotting, the operating times of known designs are often limited in order to ensure safe and reliable treatment.

[0011] Therefore, there is a need for providing an apparatus for warming fluids and a corresponding container for fluids that reduces or minimizes air retention and/or the risk of blood clotting in case blood is conveyed through the container.

SUMMARY

[0012] A general aim of the present invention is to provide an apparatus for controlling the temperature of fluids for use in an extracorporeal blood treatment apparatus that alleviates or minimizes the above-mentioned drawbacks.

[0013] It is a further aim of the present invention to provide a container suitable for use with an apparatus for controlling the temperature of (e.g., warming or cooling) fluids that alleviates or minimizes the above-mentioned drawbacks.

[0014] It is a further aim of the present invention to provide a container for use with an apparatus for controlling the temperature of (e.g., warming or cooling) fluids that minimizes air retention and/or the risk of blood clotting by preventing or minimizing the formation of low fluid flow areas in an internal conduit of the container.

[0015] It is a further aim of the present invention to provide a container for use with an apparatus for controlling the temperature of (e.g., warming or cooling) fluids that minimizes pressure drop and/or risk of hemolysis by preventing or minimizing the formation of high fluid flow areas in an internal conduit of the container.

[0016] It is a further aim of the present invention to provide a container for use with an apparatus for controlling the temperature of (e.g., warming or cooling) fluids that minimizes air retention and/or the risk of blood clotting especially at lower fluid flow rates through the container. This may also entail substantial advantages in applications in which fluids other than blood are conveyed through the container.

[0017] It is a further aim of the present invention to provide a container for use with an apparatus for warming fluids that minimizes the risk for formation of hot spots by preventing or minimizing the retention of air bubbles in an internal conduit of the container. This may also entail preventing damage to the fluid (e.g. hemolysis in the case of blood, degassing or chemical decomposition in other cases) and/or to the material the container is made of, for example, due to overheating.

[0018] It is a further aim of the present invention to provide a container for use with an apparatus for warming fluids that improves or maximizes heat transfer by preventing the formation of hot spots and/or by minimizing the retention of air bubbles.

[0019] This may also entail improving or optimizing the heat exchange between heating surfaces to the apparatus and the fluid conveyed through the container.

[0020] It is a further aim of the present invention to provide a container for use with an apparatus for controlling the temperature of (e.g., warming or cooling) fluids that can be manufactured at the same or lower cost as known containers, while providing one or more of the above advantages.

[0021] At least one of the above-indicated aims is attained by a container and an apparatus according to one or more of the appended claims, taken singly or in any combination.

[0022] In a 1.^{sup}.st independent aspect there is provided a container for fluids, in particular a container for warming or cooling fluids, the container comprising an inlet port; an outlet port; and a fluid conduit configured for putting the inlet port in fluid communication with the outlet port and comprising one or more deflection sections. The fluid conduit has a non-constant maximum width in a direction of fluid flow F through the fluid conduit. At least one of the one or more deflection sections further comprises an entry section and an exit section, each respective exit section being arranged downstream, in the direction of fluid flow, from each respective entry section.

[0023] The maximum width of the fluid conduit decreases along the direction of fluid flow through the entry section over a first distance $D_{sub.e}$ and the maximum width of the fluid conduit increases along the direction of fluid flow through the exit section over a second distance $D_{sub.x}$, the first distance and the second distance being different from one another.

[0024] In a 2.^{sup}.nd aspect according to the 1st aspect, the direction of fluid flow F through the fluid conduit is based on a direction from the inlet port towards the outlet port.

[0025] In a 3.^{sup}.rd aspect according to anyone of the preceding aspects, the first distance $D_{sub.e}$ is smaller than the second distance $D_{sub.x}$.

[0026] In a 4.^{sup}.th aspect according to anyone of the preceding aspects, the maximum width of the fluid conduit decreases along the direction of fluid flow F through the entry section from a first width $L_{sub.c}$ to a second width $L_{sub.t}$.

[0027] In a 5.^{sup}.th aspect according to the preceding aspect, with

$$[00001] 0.5 \leq \frac{L_t}{L_c} \leq 0.85,$$

particularly with

$$[00002] 0.7 \leq \frac{L_t}{L_c} \leq 0.8,$$

more particularly with

$$[00003] \frac{L_t}{L_c} = 0.75.$$

[0028] In a 6.^{sup}.th aspect according to anyone of the two preceding aspects, the maximum width of the fluid conduit increases along the direction of fluid flow F through the exit section from the second width $L_{sub.t}$ to the first width $L_{sub.c}$.

[0029] In a 7.^{sup}.th aspect according to anyone of the preceding aspects, the maximum width of the fluid conduit decreases non uniformly along the direction of fluid flow F through the entry

section with at least a first phase and a second phase, the decrease in the first phase and the decrease in the second phases being different from one another.

[0030] In an 8.sup.th aspect according to anyone of aspects 1 to 6, the maximum width of the fluid conduit decreases along the direction of fluid flow F through the entry section less than linearly in a first phase and more than linearly in a second phase.

[0031] In a 9.sup.th aspect according to anyone of the two preceding aspects, the first phase covers at least 50% of the first distance D.sub.e.

[0032] In a 10.sup.th aspect according to anyone of the preceding aspects, the maximum width of the fluid conduit increases along the direction of fluid flow F through the exit section substantially constantly over the second distance D.sub.x.

[0033] In an 11.sup.th aspect according to anyone of the preceding aspects, the first distance D.sub.e is between about 0.3 times the maximum width and about 2.0 times the maximum width, particularly between about 0.5 times the maximum width and about 0.7 times the maximum width, more in detail about 0.6 times the maximum width.

[0034] In a 12.sup.th aspect according to anyone of the preceding aspects, the second distance D.sub.x is between about 0.8 times the maximum width and about 4.0 times the maximum width, particularly between about 1.0 times the maximum width and about 2.0 times the maximum width.

[0035] In a 13.sup.th aspect according to anyone of the preceding aspects, for each of the one or more deflection sections, the entry section has a first end and a second end, the first end of the entry section being upstream, in the direction of fluid flow F, of the second end of the entry section; and the exit section has a first end and a second end, the first end of the exit section being upstream, in the direction of fluid flow F, of the second end of the exit section.

[0036] In a 14.sup.th aspect according to the preceding aspect, the maximum width of the fluid conduit at the first end of the entry section is substantially equal to the maximum width of the fluid conduit at the second end of the exit section.

[0037] In a 15.sup.th aspect according to anyone of the two preceding aspects, the maximum width of the fluid conduit at the first end of the entry section is between about 18 mm and 22 mm, particularly between about 19 mm and about 21 mm, and/or the maximum width of the fluid conduit at the second end of the entry section is between about 13 mm and 17 mm, particularly between about 14 mm and about 16 mm.

[0038] In a 16.sup.th aspect according to anyone of the three preceding aspects 1 to 14, the maximum width of the fluid conduit at the first end of the entry section is between about 38 mm and 42 mm, particularly between about 39 mm and about 41 mm, and/or the maximum width of the fluid conduit at the second end of the entry section is between about 28 mm and 32 mm, particularly between about 29 mm and about 31 mm.

[0039] In a 17.sup.th aspect according to anyone of the preceding aspects 13 to 15, the maximum width of the fluid conduit at the second end of the exit section is between about 18 mm and 22 mm, particularly between about 19 mm and about 21 mm, and/or the maximum width of the fluid conduit at the first end of the exit section is between about 13 mm and 17 mm, particularly between about 14 mm and about 16 mm.

[0040] In an 18.sup.th aspect according to anyone of the preceding aspects 13, 14, 16, the maximum width of the fluid conduit at the second end of the exit section is between about 38 mm and 42 mm, particularly between about 39 mm and about 41 mm, and/or the maximum width of the fluid conduit at the first end of the exit section is between about 28 mm and 32 mm, particularly between about 29 mm and about 31 mm.

[0041] In a 19.sup.th aspect according to anyone of the preceding aspects, each of the one or more deflection sections further comprises an intermediate section interposed between the entry section and the exit section.

[0042] In a 20.sup.th aspect according to the preceding aspect, each respective intermediate section has a substantially constant width.

[0043] In a 21.sup.st aspect according to anyone of aspects 4 to 6, in combination with the preceding aspect, the constant width is equal to the second width L.sub.t.

[0044] In a 22.sup.nd aspect according to anyone of the three preceding aspects, each respective intermediate section is directly adjacent to the corresponding entry section, particularly the corresponding entry section being a direct extension of the respective intermediate section.

[0045] In a 23.sup.rd aspect according to anyone of the four preceding aspects, each respective intermediate section is directly adjacent to the corresponding exit section, particularly the corresponding exit section being a direct extension of the respective intermediate section.

[0046] In a 24.sup.th aspect according to anyone of the five preceding aspects, the intermediate section is provided with an inner radius R.sub.2 calculated as

$$[00004] R_2 = (1 - \frac{L_t}{L_c}) \cdot \text{Math. } L_c + \frac{\text{weld}}{2},$$

with L.sub.c=the maximum width, L.sub.t=the width of the intermediate section (110i), and weld=width of weld; and/or the ratio

$$[00005] \frac{R_2}{L_c}$$

between the inner radius (R.sub.2) and the maximum width (L.sub.c) ranges from 0.15 to 0.50, particularly from 0.2 to 0.3, and is more in particular equal to about 0.25.

[0047] In a 25.sup.th aspect according to anyone of the six preceding aspects, the intermediate section is provided with an outer radius R.sub.3 calculated as

$$[00006] R_3 = L_c + \frac{\text{weld}}{2},$$

with L.sub.c=the maximum width, and weld=width of weld.

[0048] In a 26.sup.th aspect according to anyone of aspects 19 to 25, the intermediate section has an inner edge and an opposite outer edge, the inner edge having a radius smaller than a radius of the outer edge. Each of the entry section and the exit section has a respective inner edge in extension to the inner edge of the intermediate section. Each of the entry section and the exit section has a respective outer edge in extension to the outer edge of the intermediate section.

[0049] In a 27.sup.th aspect according to the preceding aspect, the maximum width of the fluid conduit decreases along the direction of fluid flow F through the entry section substantially due to a directional change of the inner edge of the entry section, optionally the outer edge of the entry section continuing substantially straight and/or tangentially in extension from the outer edge of the intermediate section.

[0050] In a 28.sup.th aspect according to anyone of the two preceding aspects, the maximum width of the fluid conduit increases along the direction of fluid flow F through the exit section substantially due to a directional change of the inner edge of the exit section, optionally the outer edge of the exit section continuing substantially straight and/or tangentially in extension from the outer edge of the intermediate section.

[0051] In a 29.sup.th aspect according to anyone of aspects 1 to 25, the maximum width of the fluid conduit decreases along the direction of fluid flow F through the entry section substantially due to a directional change of an inner edge of the entry section, and/or an outer edge of the entry section is substantially straight.

[0052] In a 30.sup.th aspect according to anyone of aspects 1 to 25 or 29, the maximum width of the fluid conduit increases along the direction of fluid flow F through the exit section substantially due to a directional change of an inner edge of the exit section, and/or an outer edge of the exit section is substantially straight.

[0053] In a 31.sup.st aspect according to anyone of the preceding aspects in combination with aspect 19, the intermediate section is provided with a deflection of at least about 90°, particularly about 180°.

[0054] In a 32.sup.nd aspect according to anyone of the preceding aspects, the fluid conduit further comprises a plurality of connection sections.

[0055] In a 33.sup.rd aspect according to the preceding aspect, the fluid conduit is provided, along

each of the plurality of connection sections, with a substantially constant maximum width; and/or the fluid conduit is, along each of the plurality of connection sections, substantially straight.

[0056] In a 34.sup.th aspect according to anyone of the two preceding aspects, the fluid conduit is provided with a maximum width along one of the plurality of connection sections equal to a maximum width along each other of said plurality of connection sections.

[0057] In a 35.sup.th aspect according to anyone of the three preceding aspects, the plurality of connection sections comprises an inlet section connected to the inlet port and to an adjacent first connection section of the plurality of connection sections, the inlet section being configured to provide the fluid conduit with a transition from a diameter of the inlet port to a maximum width of the fluid conduit at the first connection section, particularly the inlet section including an inner edge and an outer edge, the inner edge and the outer edge each forming a respective inlet angle with respect to an axis of the inlet port of about 5° to about 30°, more particularly of less than about 15°.

[0058] In a 36.sup.th aspect according to anyone of the four preceding aspects, the plurality of connection sections comprises an outlet section connected to the outlet port and to an adjacent second connection section of the plurality of connection sections, the outlet section being configured to provide the fluid conduit with a transition from a diameter of the outlet port to a maximum width of the fluid conduit at the second connection section, particularly the outlet section including an inner edge and an outer edge, the inner edge and the outer edge each forming a respective outlet angle with respect to an axis of the outlet port of about 25° to about 60°, more particularly of less than about 40°.

[0059] In a 37.sup.th aspect according to anyone of the preceding aspects, the fluid conduit has a meandering or serpentine shape.

[0060] In a 38.sup.th aspect according to anyone of the preceding aspects, the one or more deflection sections include a number of deflection sections, the number of deflection sections being uneven, optionally the number of deflection sections being equal to 1, 3, 5, 7, 9 or 13; particularly the number of deflection sections being 1 or 3.

[0061] In a 39.sup.th aspect according to the preceding aspect, an uneven number of deflection sections of the one or more deflection sections is arranged opposite to an even number of deflection sections of the one or more deflection sections.

[0062] In a 40.sup.th aspect according to anyone of the preceding aspects, the one or more deflection sections include at least three deflection sections.

[0063] In a 41.sup.th aspect according to the preceding aspect, at least one of the at least three deflection sections is arranged opposite to at least two of the at least three deflection sections.

[0064] In a 42.sup.th aspect according to anyone of the preceding aspects, the container further comprises a proximal end and a distal end opposite the proximal end.

[0065] In a 43.sup.th aspect according to the preceding aspect, both the inlet port and the outlet port are arranged at the proximal end.

[0066] In a 44.sup.th aspect according to anyone of the two preceding aspects, the container is configured to be inserted, for use, with the distal end first into a receptacle of an apparatus for warming or cooling fluids.

[0067] In a 45.sup.th aspect according to anyone of the preceding aspects, the inlet port is configured for connecting to a fluid inlet line of a blood treatment apparatus.

[0068] In a 46.sup.th aspect according to the preceding aspect, the inlet port is configured for receiving medical fluid from the fluid inlet line through the inlet port.

[0069] In a 47.sup.th aspect according to anyone of the preceding aspects, the outlet port is configured for connecting to a fluid outlet line of a blood treatment apparatus.

[0070] In a 48.sup.th aspect according to the preceding aspect, the outlet port is configured for releasing medical fluid from the outlet port into the fluid outlet line.

[0071] In a 49.sup.th aspect according to anyone of the preceding aspects, the container is made

from a substantially flexible material, optionally the material including one or more of polyurethane (PUR) and polyvinylchloride (PVC).

[0072] In a 50.sup.th aspect according to anyone of the preceding aspects, the container comprises a bag, the bag optionally comprising at least a first and a second film, the first and the second films being sealed to one another.

[0073] In a 51.sup.st aspect according to anyone of the preceding aspects, particularly during use when liquid is going through the conduit, the conduit exhibits a ratio between the maximum width and a maximum inner height (w/h) greater than 5, in particular greater than 10.

[0074] In a 52.sup.nd aspect according to anyone of the preceding aspects, during use, the fluid flow inside the conduit is a laminar flow.

[0075] In a 53.sup.rd independent aspect, there is provided an apparatus for warming or cooling medical fluids, comprising a receptacle; a heating region; a container for warming or cooling medical fluids according to any one of aspects 1 to 52 received in the receptacle; and the heating region is configured to contact at least a portion of the container for transferring heating energy to a fluid flowing through the container.

[0076] In a 54.sup.th aspect according to the preceding aspect, the heating region further comprises a first heating surface and optionally a second heating surface, the first and optional second heating surfaces being configured to contact first and second surfaces of the container for transferring heating energy to a fluid flowing through the container via the first surface and optionally the second surface.

[0077] In a 55.sup.th aspect according to anyone of the two preceding aspects, the receptacle further comprises an extension configured to receive a matching portion of the bag in order to ensure one or more of: a correct placement of the container in the receptacle, a correct orientation of the container when placed in the receptacle, a correct positioning of the container with respect to the apparatus for warming or cooling fluids.

[0078] In a 56.sup.th aspect according to anyone of the three preceding aspects, the receptacle further comprises, alternatively or in combination a reference protrusion configured to engage a matching reference opening of the bag; and/or a sensor, e.g. an optical sensor, configured to sense the presence of a matching portion of the bag in an extension of the receptacle; in order to ensure one or more of a correct placement of the container in the receptacle, a correct orientation of the container when placed in the receptacle, a correct positioning of the container with respect to the apparatus for warming or cooling fluids.

[0079] In a 57.sup.th independent aspect there is provided an extracorporeal blood circuit comprising a blood withdrawal line connectable to an inlet of a primary chamber of a filtration unit; a blood return line connected to an outlet of the primary chamber, the blood lines being configured for connection to a cardiovascular system of a patient; optionally a dialysis effluent line connected to an outlet of a secondary chamber of the filtration unit. The extracorporeal blood circuit further comprises a container for warming or cooling fluids according to anyone of aspects 1 to 52 connected to the blood return line or to the blood withdrawal line.

[0080] In a 58.sup.th independent aspect there is provided an apparatus for extracorporeal blood treatment comprising a filtration unit having a primary chamber and a secondary chamber separated by a semi-permeable membrane; a blood withdrawal line connected to an inlet of the primary chamber; a blood return line connected to an outlet of the primary chamber, the blood lines being configured for connection to a cardiovascular system of a patient; optionally a dialysis supply line connected to an inlet of the secondary chamber; optionally a dialysis effluent line connected to an outlet of the secondary chamber; and at least one pump to move the fluids in the lines; and a control unit driving the pump. The apparatus further comprises an apparatus for warming or cooling medical fluid according to any one of aspects 53 to 56.

[0081] In a 59.sup.th independent aspect, optionally according to anyone of the previous aspects, there is provided a container **100** for fluids comprising: an inlet port **112**, inlet tubing **111** having a

section diameter and being connected to the inlet port **112**; an outlet port **116**; and a fluid conduit configured for putting the inlet port in fluid communication with the outlet port and comprising one or more deflection sections **110**, wherein the fluid conduit has a non-constant maximum width $L_{sub.c}$ in a direction of fluid flow F through the fluid conduit and a height h , the section diameter of the inlet tubing being larger than the height h of the fluid conduit; at least one of the one or more deflection sections further comprises an entry section **110e** and an exit section **110x**, each respective exit section being arranged downstream, in the direction of fluid flow, from each respective entry section; an inlet section **113** fluidly connecting the inlet tubing **111** with the fluid conduit, wherein the inlet section **113** comprises a first plane and a second plane, opposite to the first plane, developing from the inlet tubing **111** towards the fluid conduit and converging towards the fluid conduit.

[0082] In a 60.sup.th aspect according to the preceding aspect, the first plane and/or the second plane of the inlet section defines an angle with respect to a longitudinal extension of a fluid conduit main plane less than 20° , particularly included between 2° and 17° .

[0083] In a 61.sup.st aspect according to the preceding two aspects, the inlet section further includes a terminal portion connecting the first and second plane with the fluid conduit, the terminal portion including two respective layers substantially parallel at a distance substantially corresponding to the height h of the fluid conduit, the first plane and the second plane being connected to a respective layer of the terminal portion.

[0084] Further characteristics and advantages of the present invention will better emerge from the detailed description that follows of at least an embodiment of the invention, illustrated by way of non-limiting example in the accompanying figures of the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0085] The description will now follow, with reference to the appended figures, provided by way of non-limiting example, in which:

[0086] FIG. **1** schematically represents an extracorporeal blood treatment apparatus in accordance with an illustrating embodiment.

[0087] FIG. **2** shows, schematically and in partially exploded view, an apparatus for warming fluids in accordance with embodiments of the present invention;

[0088] FIG. **3** shows a bag for warming fluids in accordance with a prior art design;

[0089] FIG. **4** shows a bag for an apparatus for warming fluids, the bag being in accordance with a first embodiment of the present invention;

[0090] FIG. **5** shows a deflection section of one or more deflection sections as used in a bag for an apparatus for warming fluids, the bag being in accordance with a second embodiment of the present invention;

[0091] FIG. **6A** shows a side view of an inlet section in accordance with embodiments of the present invention;

[0092] FIG. **6B** shows a perspective view of an inlet section in accordance with embodiments of the present invention;

[0093] FIGS. **7A** to **7D** show fluid flow based on different alternative embodiments of inlet sections in accordance with embodiments of the present invention;

[0094] FIG. **8A** shows a bag for an apparatus for warming fluids, the bag being in accordance with another embodiment of the present invention; and

[0095] FIGS. **8B** and **8C** shows details of the bag embodiment of FIG. **8A**.

DETAILED DESCRIPTION

[0096] FIG. **1** schematically represents an extracorporeal blood treatment apparatus **1** in

accordance with an illustrating embodiment.

[0097] An example of an extracorporeal blood treatment circuit **200** is schematically illustrated, but it is noted that the specific structure of the extracorporeal blood treatment circuit **200** is not relevant for the purposes of the present invention and therefore other and different circuits to those specifically shown in FIG. **1** might be used in consequence of the functional and design needs of each single medical apparatus (e.g. continuous renal replacement treatments—CRRT treatments—vs chronic dialysis treatments).

[0098] The extracorporeal blood treatment circuit **200** exhibits a dialysis fluid circuit **32** having a dialysis fluid supply line **8**, configured to transport a dialysis liquid from at least one source **14** towards a treatment station **15** where one or more filtration units **2**, or dialyzers, operate.

[0099] The dialysis fluid circuit **32** further comprises a dialysis effluent line **13**, configured for the transport of a dialysate liquid (spent dialysate and liquid ultrafiltered from the blood through a semipermeable membrane **5**) from the treatment station **15** towards an evacuation zone, schematically denoted by **16** in FIG. **1**.

[0100] The hydraulic circuit cooperates with a blood circuit **17**, also schematically represented in FIG. **1** in its basic component parts. The specific structure of the blood circuit is also not fundamental to the present invention. Thus, with reference to FIG. **1**, a brief description of a possible embodiment of a blood circuit is made, which is however provided purely by way of non-limiting example.

[0101] The blood circuit **17** of FIG. **1** comprises a blood withdrawal line **6** configured to remove blood from a vascular access **18** and a blood return line **7** configured to return the treated blood to the vascular access **18**.

[0102] The blood circuit **17** of FIG. **1** further comprises a primary chamber **3**, or blood chamber, of the blood filtration unit **2**, the secondary chamber **4** of which is connected to the extracorporeal blood treatment circuit **200**.

[0103] In greater detail, the blood withdrawal line **6** is connected to the inlet of the primary chamber **3**, while the blood return line **7** is connected to the outlet of the primary chamber **3**.

[0104] In turn, the dialysis supply line **8** is connected to the inlet of the secondary chamber **4**, while the dialysis effluent line **13** is connected to the outlet of the secondary chamber **4**.

[0105] The filtration unit **2**, for example a dialyzer, a plasma filter, a hemofilter, or a hemodiafilter, comprises, as mentioned, the two chambers **3** and **4**, which are separated by a semipermeable membrane **5**, for example of the hollow-fiber type or of the plate type.

[0106] In an embodiment, the filtration unit (**2**) may include an adsorption device, such as a plasma filtration adsorption device, a charcoal column, an adsorption device for endotoxin removal for e.g. sepsis treatment; in this embodiment, both a fresh dialysis fluid line and a dialysis effluent line to remove spent dialysis fluid may not be present.

[0107] The blood circuit **17** may also comprise one or more air separators **19** and clamps **20** on both withdrawal and return line.

[0108] The extracorporeal blood treatment apparatus **1** comprises one or more blood pumps **21**, for example positive displacement pumps such as peristaltic pumps; in the example of FIG. **1**, a blood pump **21** is included on the blood withdrawal line **6**.

[0109] The apparatus of the above-described embodiment may also comprise a user interface **22** (e.g. a graphic user interface or GUI) and a control unit **12**, i.e. a programmed/programmable control unit, connected to the user interface.

[0110] A bypass line **23** connects the dialysis fluid supply line **8** and the dialysis effluent line **13**, thereby bypassing the filtration unit **2**, and one or more fluid check members **24** connected to the control unit **12** selectively opens and closes the bypass line **23**.

[0111] A dialysis fluid pump **25** and a dialysate pump **26** may be included, located respectively on the dialysis fluid supply line **8** and on the dialysis effluent line **13** and further operatively connected to the control unit **12**.

[0112] The apparatus may also comprise a dialysis fluid source such as one or more bags of fresh fluid (e.g. in CRRT treatment) or a dialysis fluid preparation device **9** (e.g. in chronic treatment), which may be of any known type, for example including one or more concentrate sources **27**, **28** and respective concentrate pumps **29**, **30** (regulating means **10**) for the delivery, as well as at least a conductivity sensor **35**.

[0113] Due to the dialysis apparatus potentially comprising various liquid sources (e.g. one or more water sources **14**, one or more concentrate sources **27**, **28**, one or more sources **33** of disinfectant liquids) connected to the dialysis supply line **8** with respective delivery lines **36**, **37**, **38** and **40**, the apparatus may exhibit, at each delivery line, a respective check member (not all are shown in FIG. **1**) and, for example, comprising a valve member **31** and **34**.

[0114] Arranged in the dialysis supply line **8**, in the direction in which the liquid circulates, there are the first flow meter **41** and the dialysis fluid pump **25**.

[0115] The dialysis effluent line **13** may be provided with a dialysate pump **26** and a second flow meter **42**. The first and second flow meters **41**, **42** may be used to control (in a known manner) the fluid balance of a patient connected to the blood circuit **17** during a dialysis session.

[0116] A sensor **11** is provided on the dialysis effluent line **13**, immediately downstream the filtration unit **2**, to measure a parameter value (e.g. conductivity) of the dialysate in the dialysis effluent line **13**.

[0117] One or more infusion lines **39** may also be included, with respective infusion pumps **43** or flow regulation valves, the infusion lines being connected up to the blood return line **7** and/or the blood withdrawal line **6** and/or directly to the patient. The liquid sources for the infusion lines may be pre-packaged bags **44** and/or liquids prepared by the apparatus itself. The infusion line **39** may either receive infusion liquid from a pre-packaged bag **44** (solid line **45a**) or from an online preparation through-branch **45b** (dotted line).

[0118] As already mentioned, the described embodiments are intended to be non-limiting examples.

[0119] In FIG. **1**, the blood return line **7** may be provided with a bag **100** for circulating fluids therein (a bag embodiment is represented in FIG. **4**) before returning blood to the patient.

[0120] The extracorporeal blood treatment machine is provided with an apparatus **300** for e.g. warming fluids circulating in the bag in order to regulate a temperature of the blood returned to the patient to a desired temperature region, for example around 37° C.

[0121] In the following description, reference is made to a fluid warming apparatus to regulate blood temperature and to a corresponding warming bag. However, the description should not be interpreted in a limitative way in this respect. An apparatus for temperature fluid control is included in the scope of the appended claims. A bag for either warming or cooling is also encompassed by the present description and claims.

[0122] Fluid warming apparatus **300** is generally designed to regulate the temperature of (e.g. to warm) medical fluids such as blood or infusion/substitution liquid. Fluid warming apparatus **300** may be employed to warm medical fluids other than blood intended to be returned or fluids intended to be supplied to the body of a patient, for example when regulating the temperature of treatment solution entering filtration unit **2**. In other terms, the bag **100** may be differently connected to the dialysis line **8**. Thus, fluid warming apparatus **300** may be employed in a manner other than that illustrated in FIG. **1**. In the illustrative embodiment shown in FIG. **1**, fluid warming apparatus **300** is employed to warm the blood being returned to the patient and coming from air separator **19**. An air detector **46** and a clamp **20** may be arranged on blood line **7** downstream from the fluid warming apparatus **300** and upstream from vascular access **18**.

[0123] Given the above description of a possible embodiment of extracorporeal blood treatment apparatus, thereafter specific embodiments of the apparatus **300** for e.g. warming fluids and of a bag **100** for fluids are described.

[0124] FIG. **2** shows, schematically and in partially exploded view, an apparatus **300** for warming

fluids in accordance with embodiments of the present invention. Fluid warming apparatus **300** may generally be designed as shown, namely configured to e.g. substantially horizontally receive, in a slot **306**, a bag **100** for e.g. warming fluids between a first **302** and a second **304** component, defining a heating region **308** between the first **302** and second **304** components. The apparatus **300** may be designed with first **302** and second **304** components pivotably connected to one another, thereby facilitating cleaning of heating region **308**. First **302** and second **304** components may, however, also be connected to one another in any other suitable manner (e.g. disconnection, shifting, tilting), thereby facilitating cleaning and/or insertion/removal of bag **100**. In the embodiment shown, first **302** and second **304** components are shown separated from one another for clarity. During use, first **302** and second **304** components are fixedly mounted to one another, thereby providing slot **306** for insertion/removal of bag **100**. As shown, a bag **100** may be inserted into slot **306** and, thus, positioned in superimposition with the heating region **308**. Slot **306** may be provided with an extension **307** in order to receive a matching portion **107** of bag **100**. Portion **107** may be in the form of a small tab protruding from one side of the bag **100** (see FIG. **8a**) or triangular shaped as shown in FIGS. **2** and **4**. Further, extension **307** of slot **306** may be provided with a reference protrusion **307r** configured to engage a corresponding reference opening **107r** in bag **100**. Tab, extension **307** and/or reference protrusion **307r** are designed to ensure proper placement of bag **100** into slot **306** as described with respect to bag **100** further below.

[0125] The heating region **308** typically comprises heating surfaces between which, during use, a bag **100** is positioned, having a large portion of its outer surfaces in contact with the heating surfaces. As shown in FIG. **2**, a bag **100** would present substantially flat upper **108-1** and lower **108-2** surfaces that may be brought into superimposition with corresponding upper **308-1** and lower **308-2** (e.g. upper and lower) heating surfaces of heating region **308**.

[0126] FIG. **3** shows a bag **100k** for warming fluids in accordance with a prior art design. The bag **100k** may generally be used with an apparatus similar or identical to apparatus **300** for warming fluids as shown in FIG. **2** and as described above.

[0127] According to the prior art design, bag **100k** is made from two adjacent layers of film material, for example plastic film, that are welded to one another in a specific manner in order to form a conduit for fluid to be heated. The conduit puts an inlet port **112k** into fluid communication with an outlet port **116k**. The conduit further includes an inlet section **113k** adjacent to inlet port **112k** and an outlet section **117k** adjacent to outlet port **116k**. As shown in FIG. **3**, the conduit further includes three deflection sections **110k**, connecting the elongated sections of bag **100k** with one another and, respectively, with the inlet section **113k** and the outlet section **117k**, providing the conduit with a generally meandering shape.

[0128] As shown in FIG. **3**, inlet section **113k**, deflection sections **110k**, outlet section **117k**, and elongated sections in between typically have substantially the same cross-section, with little or no variation in the width and/or cross-section of the conduit over any of the sections.

[0129] The direction of fluid flow **F** is shown in FIG. **3** for each of the elongated sections and is generally directed from the inlet port **112k** towards the outlet port **116k**. As illustrated, fluid flowing through the conduit from inlet port **112k** towards outlet port **116k** enters the conduit at the inlet section **113k** and exits the conduit at outlet section **117k**. The fluid is also deflected according to the meandering shape of the conduit at the deflection sections **110k**. The fluid flow profile through the conduit is dependent on the geometry of the conduit in the respective sections (e.g. inlet section, outlet section, elongated section(s), deflection section(s)), flow rate and viscosity. While the fluid flow is relatively fully developed over the elongated sections, it varies with respect to a number of fluid flow parameters throughout the sections of the conduit.

[0130] As can be seen from FIG. **3**, fluid flow through the conduit exhibits regions **A** of fluid flow having a relatively lower velocity or even stagnating, regions **B** of relatively homogeneous fluid flow, and regions **C** of fluid flow having relatively higher velocity.

[0131] Low velocity fluid flow may entail substantial disadvantages, for example areas with little

or no exchange of fluid due to the formation of stagnation areas and/or to the collection or accumulation of air bubbles in areas of low fluid flow. If the fluid conveyed through the conduit is blood, low velocity fluid flow may lead to clotting, entailing the risk of blood clots being created, being carried away by the blood flow, and ultimately being returned to the patient in the blood flow.

[0132] This is particularly likely in case of stagnation of fluid, when regions are formed in which little or no fluid flow occurs and where little to none exchange of fluid with the mass of fluid flowing through the conduit takes place. Low velocity fluid flow is, thus, to be avoided.

[0133] Low velocity fluid flow, denoted in FIG. 3 as regions A, may occur at the inlet section **113k**, at the outlet section **117k**, and at the deflection section **110k**. The shape of the deflection sections **110k** may significantly contribute to the formation of low velocity fluid flow, in particular in the intermediate section thereof (see region A along an outer edge of the intermediate section) and in the exit section thereof (see region A located at an inner edge of the exit section, downstream from the intermediate section).

[0134] High velocity fluid flow may also entail disadvantages, for example excessive pressure drop between the inlet and the outlet of the container and high shear stress in the near container wall vicinity.

[0135] High velocity fluid flow, denoted in FIG. 3 as regions C, may occur near the inlet section **113k**, near the outlet section **117k**, and at the deflection section **110k**. High velocity fluid flow, occurring near the inlet section **113k** or near the outlet section **117k** is typically less critical, since the higher velocity is generally caused by the fluid flowing from medical tubing through the inlet port **112k** (or towards the outlet port **116k**), which is typically provided with a smaller cross-section than that of the conduit. However, high velocity fluid flow occurring near the inlet section **113k** or near the outlet section **117k** may have detrimental effects as well.

[0136] The shape of the fluid path geometry (in combination with fluid flow rate and viscosity) causes low/high velocity regions for the fluid. High velocity fluid flow occurring at the deflection section **110k** typically occurs at an inner edge of the intermediate section thereof. Excessively high velocity fluid flow is to be avoided since it may cause blood haemolysis.

[0137] FIG. 4 shows a bag **100** for an apparatus **300** for warming fluids, the bag **100** being in accordance with a first embodiment of the present invention. FIG. 8A shows a second embodiment of bag **100** particularly designed for high flow rates. The bag **100** is flexible and in particular made of a film material.

[0138] The bag **100** may be made from two layers of film material, particularly polyurethane (PUR) or polyvinylchloride (PVC), superposed and welded to form the bag **100** and to form a conduit delimited by the two layers and by the lines of welding. Lines of welding are not shown in FIG. 4 for clarity. For illustration purposes, lines of welding **120** are shown in the detailed view of FIG. 5.

[0139] Polyurethane (PUR) is a material having high mechanical resistance and good thermal transfer properties. Therefore, the use of PUR may provide bag **100** with advantageous properties regarding heat transfer through the film material and to the fluid inside bag **100**. The use of PUR may further minimize the risk of leakage of fluid from bag **100**.

[0140] The embodiment shown in FIG. 4 is provided with four connection sections **102** arranged substantially parallel to one another in a layout exhibiting a substantially side-by-side arrangement of the four connection sections **102**.

[0141] Vice versa, the embodiment shown in FIG. 8A is provided with two connection sections **102** only. It is noted that the connection sections **102** are shown as parallel to an overall development direction of bag **100** (e.g. substantially parallel to directions F and/or the direction of fluid inflow and outflow—see arrows near element **111**). However, connection sections **102** may be arranged at an angle (e.g. orthogonal) with respect to the overall development direction and/or with respect to one another. In the latter case, the cross-section and/or maximum width $L_{sub.c}$ of the conduit may

increase or decrease to a certain extent along the length of any of the sections (e.g. connection sections, deflection sections, etc.).

[0142] The terms width and cross-section both pertain to a measure of cross sectional size of the conduit and its sections and are used to reflect the different states of bags **100** before and during use. As the bags **100** are made of layers of plastic film, a bag has, before use, a substantially flat shape, due to the conduit not containing any liquid or particles. In this unused state, the size of the conduit may be referred to as having a width, since the width of the flat conduit is an effective measure thereof. During use, however, the fluid being conveyed through the conduit causes the two layers of film to cease contacting each other, thereby opening up the conduit vertically and facilitating fluid flow. In this used state, the size of the conduit may be referred to as having a cross-section, since the cross-section (or cross section area) is an effective measure thereof. Typically, in one embodiment (FIG. **4**) the cross section of the conduit has, during use, a size of approximately 1-2 mm (height)×18-22 mm (width), in particular about 1.4 mm×20 mm. The width or maximum width $L_{sub.c}$ of the conduit, when the bag is not used (i.e. flat), is approximately 20 mm. In some embodiments (FIG. **8A**), the cross section of the conduit may be larger; the cross section of the conduit has, during use, a size of approximately 1-2 mm (height)×34-45 mm (width), in particular about for example 1.4 mm×40 mm, for high flow applications. It is noted that the main principles of the overall design, in particular those of having a conduit with a width that is much larger than its height (i.e. substantially 2D flow with width>>height) and the conduit being designed for laminar flow, may be applied in a broad range of applications and is not restricted to the dimensions given (e.g. with respect to the application for blood warmer bags). In this respect several properties may be adjusted for individual applications, for example the relation between the heat exchange surface area with respect to a target performance (e.g. $W \times L$, conduit width×conduit length), minimum/maximum shear rates in the target flow operating range (e.g. depending on width and height of the conduit), and pressure drop (e.g. depending on width, height, and length of the conduit).

[0143] In this respect when referring to 2D flow, it is intended that the fluid flows in a channel having a width much higher than the height so that, though being of course the flow three dimensional, it may be considered substantially bi-dimensional. In more detail, the ratio between the width and the height is higher than 5 ($L_{sub.c}/h > 5$), possibly higher than 10. The two examples of FIGS. **4** and **8A** have a ratio $L_{sub.c}/h$ of more than 10.

[0144] Moreover, the present bag embodiments are more advantageous in presence of, though not limited, and particularly designed for laminar flow conditions inside the bag.

[0145] Generally referring to the disclosed embodiments (FIGS. **4** and **8A**), opposite ends of the meandering conduit are arranged on the same side of bag **100** and are provided with an inlet section **113** and an outlet section **117**. The inlet section **113** and the outlet section **117** are further connected to an inlet port **112** and an outlet port **116**, respectively. The inlet **112** and outlet **116** ports may further include a tube or medical fluid line **111** and/or a Luer connector (not shown). The Luer connectors may be arranged at an opposite end of the tubes **111**. In particular tube **111** connected to the inlet **112** may be provided with a male Luer connector and tube **111** connected to the outlet **116** may be provided with a female Luer connector so to avoid erroneous connection of the bag **100** to the extracorporeal blood circuit. In some embodiments, the Luer connectors may be arranged adjacent to inlet port **112** and/or outlet port **116**, respectively. The tubes may also be made from, for example, PVC or PUR.

[0146] More generally, the inlet **112** is provided with a connector configured to be coupled with a respective counter connector on the return line **7** placed downstream the filtration unit **2** and possibly downstream the air separator **19**, but immediately upstream the warmer unit **300** and upstream the air detector **46**; the outlet **116** is provided with a connector configured to be coupled with a respective counter connector on the return line **7** placed immediately downstream the warmer unit **300** and upstream the air detector **46**; the inlet and outlet connectors and the counter

connectors are configured to be coupled only in the correct configuration (i.e. the inlet connector may not be coupled to the outlet counter-connector and the outlet connector may not be coupled to the inlet counter-connector).

[0147] Adjacent connection sections **102** are joined together and put in fluid communication by means of deflection sections **110**. Deflection sections **110** define rounded bends that provide the conduit with an overall meandering (or serpentine) shape. Each deflection section **110** includes a respective entry section **110e**, a respective intermediate section **110i**, and a respective exit section **110x**, arranged in sequence based on fluid flow through the conduit (see arrows F in FIGS. **4** and **8A**).

[0148] As shown in FIG. **4** and FIG. **8A**, the entry section **110e**, the intermediate section **110i**, and the exit section **110x** of each deflection section **110** are provided with a specific shape in order to avoid one or more of the problems mentioned above associated with prior art designs. In the embodiments shown, the entry section **110e** provides the conduit with a decrease in the maximum width (or internal diameter) from the adjacent connection section **102** (having a larger width/diameter) towards the adjacent intermediate section (having a smaller width/diameter). The exit section **110x** provides the conduit with an increase in the maximum width (or internal diameter) from the adjacent intermediate section (having a smaller width/diameter) towards the adjacent connection section **102** (having a larger width/diameter). Intermediate section **110i** is shown as having a substantially constant width/diameter.

[0149] The individual shapes of entry section **110e** and exit section **110x** are different from one another. Entry section **110e** includes an inner edge **110e-1** and an outer edge **110e-2**, inner and outer being defined with respect to the corresponding intermediate section **110i**, which defines an outer edge **110i-2** (i.e. the outside of the bend) and an inner edge **110i-1** (i.e. the inside of the bend). The inner edge **110e-1** of entry section **110e** is provided with a nonlinear shape (e.g. corresponding substantially to a segment of a circle) while the outer edge **110e-2** of entry section **110e** is provided with a straight shape (e.g. continuing straight and in extension from the corresponding outer edge **102-2** of the preceding connection section **102**). The nonlinear shape of the inner edge **110e-1** of entry section **110e** determines the decrease in width/cross-section of the conduit from the preceding connection section **102** to the following intermediate section **110i**.

[0150] With respect to the terms “inner edge” and “outer edge” in connection with connection sections **102**, the following is noted. For connection sections **102**, which are provided to connect deflection sections **110** with one another, the terms “inner” and “outer” with respect to the edges of a connection section **102** pertain to the respective nearest deflection section **110**. FIG. **4** shows this naming convention for reasons of clarity only for one of the two connection sections **102** provided for connection deflection sections **110** with one another (see the second and third connection section **102** in direction of fluid flow F). In direction of fluid flow F, the third connection section **102** is provided with reference numerals showing the respective inner and outer edges thereof. Thus, the top edge (i.e. the edge closer to the top of FIG. **4**) of the third connection section **102** is marked as inner edge **102-1** near the second deflection section **110** (in direction of fluid flow F, and adjacent to identification region **109**). The bottom edge (i.e. the edge closer to the bottom of FIG. **4**) of the third connection section **102** is marked, along the same section of the conduit, as outer edge **102-2** near the second deflection section **110**.

[0151] Correspondingly, the top edge of the third connection section **102** is marked as outer edge **102-2** near the third deflection section **110** (in direction of fluid flow F, and adjacent to portion **107**). The bottom edge of the third connection section **102** is marked, along the same section of the conduit, as inner edge **102-1** near the third deflection section **110**. In this manner, the terms “inner” and “outer” always correspond to the nearest deflection section **110**. Therefore, the terms “inner” and “outer” change in direction of fluid flow F along the conduit depending on the nearest deflection section **102**, thereby clearly identifying the portions of connection sections **102** which are referred to as inner **102-1** and outer **102-2** edges.

[0152] It is noted that the above also applies to the remaining connection sections **102**, which are provided to connect deflection sections **110** to one another (i.e. it applies also to the second connection section **102** shown in FIG. 4). Not all reference numerals have been added to FIG. 4, however, for reasons of clarity. It is further noted that the above also applies in embodiments having fewer or more connection sections **102** than the embodiment shown in FIG. 4 (e.g. having 2 or 6 or more connection sections **102**).

[0153] In both embodiments of FIGS. 4 and 8A, similar to entry section **110e**, exit section **110x** includes an inner edge **110x-1** and an outer edge **110x-2**, inner and outer being defined with respect to the corresponding intermediate section **110i** in the same manner as described with respect to entry section **110e** above. The inner edge **110x-1** of exit section **110x** is provided with a substantially linear shape (e.g. substantially corresponding to a line segment positioned at an angle with respect to the corresponding edge of the following connection section **102**) while the outer edge **110x-2** of exit section **110x** is provided with a straight shape (e.g. leading straight and in extension to the corresponding edge of the following connection section **102**). The angular configuration of the inner edge **110x-1** of exit section **110x** with respect to the corresponding edge of the adjacent (following) connection section **102** determines the increase in width/cross-section of the conduit from the intermediate section **110i** to the following connection section **102**.

[0154] In the embodiment shown in FIG. 4, the conduit of bag **100** includes three deflection sections **110**. All deflection sections **110** are substantially identical to one another, apart from having a mirror-inverted layout depending on the deflection section **110** determining a right turn or a left turn (in direction of fluid flow and as seen from the top; see FIG. 4). Thus, the description of entry sections **110e**, intermediate sections **110i**, and exit sections **110x** is applicable to any one of the deflection sections **110** shown in FIG. 4. It is noted that, depending upon the deflection section **110** determining a right turn or a left turn, the positions of “inner” and “outer” edges changes accordingly, so that with respect to the connection sections **102** the terms “inner” and “outer” receive their respective meaning in relation to the deflection section **110**, which is referred to in the respective context (i.e. based on a respective inlet end or outlet end of a connection section **102**).

[0155] As can be seen from FIG. 4 (and similarly for embodiment of FIG. 8A), the fluid flow through the conduit defined by bag **100** is rather homogeneous. At the intermediate sections **110i** of the deflection sections **110**, there is little to no stagnation or regions of low velocity fluid flow. This is achieved by a moderate overall increase in fluid flow velocity in the intermediate sections **110** due to the decrease in width/cross-section in the respective entry section **110e**, which is maintained throughout the intermediate section **110i**. Further, the region of high velocity fluid flow (see region C—FIG. 4) is less focused/localized and spreads across a longer and wider portion of the deflection section **110**. This reduces or minimizes shear stress and causes the high velocity fluid flow in region C to be more homogeneous.

[0156] Another effect of the individual configuration of the deflection sections **110** is that there is little to no low velocity fluid flow (and no stagnation) in the respective exit sections **110x** of the deflection sections **110**. This is achieved both by the particular configuration of the exit section **110x** and by the less focused/localized and more spread-out region C of high velocity fluid flow throughout the intermediate sections **110i**.

[0157] The ratio between the width/cross-section of the connection sections **102** and the width/cross-section of intermediate sections **110** ranges between 0.5 and 0.85, particularly between 0.7 and 0.8.

[0158] Tests have been conducted with different fluid flow rates. At lower flow rates of, for example, 100 ml/min, the effect of the individual configuration of the deflection sections **110** in line with the embodiment shown in FIG. 4 is more pronounced. However, the effect may also be significant as compared to known designs at higher fluid flow rates of, for example, 200 ml/min or 300 ml/min, at which the beneficial effects described above also continue to occur.

[0159] The bag **100** in accordance with the present invention may further include inlet **113** and

outlet **117** sections configured to improve fluid flow through the conduit. In some embodiment either one or both of inlet port **112** and outlet port **116** may be arranged excentrically with respect to a main development axis of the adjacent connection section **102**.

[0160] In the embodiment shown in FIGS. **4** and **8A**, inlet port **112** and outlet port **116** are arranged excentrically with respect to the respective connection section **102** respectively putting the inlet **113** and outlet **117** sections in fluid communication with the following/preceding deflection section **110**. In the embodiment of FIG. **4**, inlet port **112** is arranged substantially parallel to the adjacent connection section **102** and slightly (laterally) shifted towards the center of bag **100** (e.g. towards and parallel to outlet port **116**). In the embodiment of FIG. **8A**, inlet port **112** is arranged substantially parallel to the adjacent connection section **102** and slightly (laterally) shifted away from the center of bag **100** (e.g. away from and parallel to outlet port **116**).

[0161] In accordance with this placement of the inlet port **112**, inlet section **113** is provided with an asymmetrical configuration, presenting an inner edge **113-1** and an outer edge **113-2**, both positioned at an angle with respect to an axis **112c** of inlet port **112**. Angles **113B** and **113a** of the inner **113-1** and outer **113-2** edges may be different from one another or the same. In some embodiments, due to the inlet port **112** not being centered with respect to the adjacent connection section **102**, the inner **113-1** and outer **113-2** edges of inlet section **113** may have different lengths, even if angles **113B** and **113a** are the same. Inlet section **113** provides the conduit with a diverging region, increasing the width/cross-section of the conduit from the diameter of the inlet port **112** to the cross-section of the adjacent connection section **102**.

[0162] The outlet section **117** is provided with inner **117-1** and outer **117-2** edges in a similar manner as described with respect to inlet section **113** above. In the embodiment of FIG. **4**, outlet port **116** is arranged substantially parallel to the adjacent connection section **102** and slightly (laterally) shifted towards the center of bag **100** (e.g. towards and parallel to inlet port **112**). In the embodiment of FIG. **8A**, outlet port **116** is arranged substantially parallel to the adjacent connection section **102** and slightly (laterally) shifted away from the center of bag **100** (e.g. away from and parallel to inlet port **112**)

[0163] In accordance with this placement of the outlet port **112**, outlet section **117** is provided with an asymmetrical configuration, presenting an inner edge **117-1** and an outer edge **117-2**, both positioned at an angle with respect to a central axis **116c** of port **116**. The angles **117β** and **117α** of the inner **117-1** and outer **117-2** edges may be different from one another or the same. Due to the outlet port **116** not being centered with respect to the adjacent connection section **102**, inner **117-1** and outer **117-2** edges may have different lengths, even if angles **117α** and **117β** are the same. Outlet section **117** provides the conduit with a converging region, decreasing the width/cross-section of the conduit from the cross-section of the adjacent connection section **102** to the diameter of the outlet port **116**.

[0164] Additionally, the inlet **113** and outlet **117** sections are provided with different shapes due to the direction of fluid flow being different for the two sections. In other words, the diverging region of inlet section **113** has a shape different from the converging region of the outlet section **117** (see FIGS. **4** and **8C**).

[0165] In both embodiments, the inlet section **113** is provided with inner **113-1** and outer **113-2** edges that form a smaller angle with respect to the corresponding edge of the adjacent connection section **102** than corresponding angles formed by inner **117-1** and outer **117-2** edges of outlet section **117**. It was found that regions A exhibiting lower velocity fluid flow at the inlet section **113** (see corresponding inlet section **113k** in FIG. **3** for comparison) may be substantially reduced by providing the inlet section **113** with a gradually increasing (w.r.t. direction of fluid flow) width/cross-section that increases the width/cross-section over a longer distance of fluid flow through the conduit.

[0166] Likewise, regions A exhibiting lower velocity fluid flow at the outlet section **117** (see corresponding outlet section **117k** in FIG. **3** for comparison) may be substantially reduced by

providing the outlet section **117** with a gradually decreasing (w.r.t. direction of fluid flow) width/cross-section that decreases the width/cross-section over a shorter distance of fluid flow through the conduit. The reduction of regions A exhibiting lower velocity fluid flow at both the inlet **113** and outlet **117** sections was most pronounced in combination with the asymmetrical arrangement of inlet **112** and outlet port **116** as shown in FIGS. **4** and **8C**.

[0167] In some embodiments, two layers of plastic film (e.g. PUR or PVC) form a portion **107**, located on the side of bag **100** opposite to the inlet/outlet ports and adjacent to the conduit. The portion **107** may be in the form of a tab (FIG. **8A**) emerging from a lateral area on the side opposite to inlet/outlet; alternatively the portion may be optionally triangular as shown in the annexed FIG. **4** by way of non-limiting example. The portion **107** is designed to fit into a correspondingly counter-shaped portion of slot **306** of the fluid warming apparatus **300**. In some embodiments, the portion **107** is provided with a reference opening **107r**. Reference opening **107r** is designed to fit with a corresponding protrusion provided in slot **306** of the fluid warming apparatus **300**. Triangular portion **107** and/or reference opening **107r** are positioned with respect to remaining components of bag **100** in order to ensure proper placement of bag **100** into slot **306** of the fluid warming apparatus **300**. In some embodiments, bag **100** is inserted into apparatus **300** generally along insertion direction G.

[0168] Alternatively (or in combination) a sensor, e.g. an optical sensor) may be used to sense the presence of the bag **100**, i.e. to sense the presence of the (tab/triangular) portion **107**.

[0169] Proper placement includes, for example, inlet/outlet tubes engaged in corresponding recesses and the conduit being placed in superimposition with heating region **308**.

[0170] Bag **100** may further include, particularly in an identification region **109** between inlet **112** and outlet **116** ports, which is not occupied by portions of the conduit, an area configured for placing a machine- or human-readable label, indicating, for example, properties of bag **100**.

[0171] FIG. **5** and corresponding FIG. **8B** show a respective deflection section **110** of one or more deflection sections **110** as used in a bag **100** for an apparatus for warming fluids **300** as described above, the bag **100** being in accordance with an embodiment of the present invention. FIGS. **5** and **8B** illustrates possible measurements for different elements of the conduit. It is noted, however, that the measurements shown are exemplary and not intended to limit the embodiments described herein.

[0172] Welding lines **120** typically have a width of about 2 mm to about 4 mm, particularly about 3 mm. The conduit according to FIGS. **4** and **5** typically has a width of about 20 mm along the connection sections **102**. As described with respect to FIG. **4** above, entry sections **110e** of deflection sections **110** typically reduce the width of the conduit from about 20 mm to about 15 mm in the intermediate sections **110i**. To this aim, entry sections **110e** may have an inner edge **110e-1** corresponding to a segment of a circle having a radius of e.g. about 10 mm (see FIG. **5**—R.sub.1). The circle segment of the inner edge **110e-1** of the entry section **110e** then continues into the inner edge **110i-1** of the intermediate section **110i**, which is defined by a segment of a circle having a radius R.sub.2 of e.g. about 4-5 mm (see FIG. **5**). The intermediate section **110i** has a width of about 15 mm (see above) and the outer edge **110i-2** thereof is defined by a radius R.sub.3 of e.g. about 21.5 mm (see FIG. **5**). The exit section **110x** has a length in direction of fluid flow F of about 30 mm and is defined by an outer edge **110x-2** continuing tangentially from the outer edge of the intermediate section **110i** and an inner edge **110x-1** positioned at an angle $110x-1\alpha$ of e.g. about 10.260 with respect to a main development axis of the following connection section **102** (e.g. an inner/outer edge thereof). In both the first and second embodiments, the outlet diverging angle, that is the angle between the inner edge **110x-1** and the corresponding edge of the connection section **102**, is less than 25°, particularly less than 15°.

[0173] FIG. **5** illustrates the asymmetrical configuration of a deflection section **110** according to embodiments of the present invention, as well as the welding lines **120** providing bag **100** with a conduit of the corresponding shape.

[0174] The conduit according to FIGS. 8A to 8C typically has a width of about 40 mm along the connection sections **102**. As described, entry sections **110e** of deflection sections **110** typically reduce the width of the conduit from about 40 mm to about 30 mm in the intermediate sections **110i**. To this aim, entry sections **110e** may have an inner edge **110e-1** corresponding to a segment of a circle having a radius of e.g. about 10 mm (see FIG. 5, R.sub.1). The circle segment of the inner edge **110e-1** of the entry section **110e** then continues into the inner edge **110i-1** of the intermediate section **110i**, which is defined by a segment of a circle having a radius of e.g. about 6.5 mm (see FIG. 5, R.sub.2). The intermediate section **110i** has a width of about 30 mm (see above) and the outer edge **110i-2** thereof is defined by a radius of e.g. about 44.5 mm (see FIG. 5, R.sub.3). The exit section **110x** has a length in direction of fluid flow **F** of about 40 mm and is defined by an outer edge **110x-1** continuing tangentially from the outer edge of the intermediate section **110i** and an inner edge **110x-2** positioned at an angle **110x-1 α** of e.g. about 140 with respect to a main development axis of the following connection section **102** (e.g. an inner/outer edge thereof). In both the first and second embodiments, the outlet diverging angle, that is the angle between the inner edge **110x-2** and the corresponding edge of the connection section **102**, is less than 25°, particularly less than 15°.

[0175] FIG. 6A shows a side view of an inlet section **113** in accordance with embodiments of the present invention. FIG. 6A illustrates a key parameter in transitioning the diameter **112d** of the inlet port **112** (or directly of the tubing **111**) to the height **h** of the connection sections **102**, namely parameter **a** describing an angle between a plane parallel to a longitudinal extension of connection sections **102** and a connecting section adjacent to the inlet port **112** or the tubing **111**. Due to the diameter of the tubing **111** or of the inlet port **112** being larger than the height **h** of the connection sections **102**, the two opposite layers defining the sides of the inlet section have an angular configuration in which angle 'a' determines over which distance, in direction of fluid flow **F**, the substantially circular diameter **112d** of the inlet port **112** or tubing **111** is adapted to the substantially flat cross section of the connection sections **102** (e.g., having a height **h** much smaller than the maximum width **L.sub.c**). FIG. 6A illustrates an angle 'a' of about 4°, resulting in the substantially circular diameter **112d** of the inlet port **112** being adapted slowly (e.g., over substantially the entire inlet section **113**) to the substantially flat cross section of the connection sections **102** (un-pinched situation).

[0176] FIG. 6B shows a perspective view of an inlet section **113** in accordance with embodiments of the present invention. FIG. 6B illustrates an angle 'a' of about 10°, resulting in the substantially circular diameter **112d** of the inlet port **112** being adapted more quickly than as shown in FIG. 6A (e.g. over a shorter section inlet section **113**) to the substantially flat cross section of the connection sections **102**. In other terms, FIG. 6B shows a situation in which the inlet section is pinched in correspondence of the second portion **113-2** where the two surfaces become parallel. As can be seen from FIGS. 6A and 6B, the corresponding sections of the inlet section **113** (or portions thereof) depend on the value of the angle 'a' and, thus, do not have to be limited to an extension of the inlet section **113** itself. FIG. 6B shows that the inlet section **113** has one portion **113-1** in which opposing layers of material are at an angle 'a' with respect to a plane parallel to the connection sections **102** (not shown) and another portion **113-2** in which the opposing layers are substantially parallel and at a distance substantially corresponding to the height **h** of the connection sections **102**.

[0177] Generally, the angle 'a' may be modified to determine the manner in which fluid entering the container **100** at the inlet port **112** is being spread laterally to conform to the substantially flat cross section of the connection sections **102**. Here, a larger angle 'a' may lead to the fluid being spread laterally more quickly or effectively (e.g. along a shorter distance in direction of fluid flow **F**) in order to reduce or eliminate the formation of regions **A** of fluid flow having a relatively lower velocity.

[0178] FIGS. 7A to 7D show fluid flow based on different alternative embodiments of inlet sections in accordance with embodiments of the present invention. FIGS. 7A to 7D illustrate the

influence of the angle 'a' on the fluid dynamics and, in particular, on the formation of regions A and C indicating regions of fluid flow having respectively relatively lower and higher velocity (e.g. contours of the velocity magnitude). FIG. 7A shows a fluid flow simulation for angle 'a'=4°, FIG. 7B shows a fluid flow simulation for angle 'a'=5°, FIG. 7C shows a fluid flow simulation for angle 'a'=10°, and FIG. 7D shows a fluid flow simulation for angle 'a'=15°. As can be seen, each angle a and each associated configuration of the respective inlet section 113 influences the fluid flow through the inlet section and, in particular, the formation of regions A and C as described above. It is noted that the examples shown in FIGS. 7A to 7D are based on the same fluid having a specific viscosity and being conveyed at the same fluid flow rate for reasons of comparison of different configurations for inlet sections 113. The pressure drop for the examples of FIGS. 7A to 7D was substantially the same with 3133 Pa, 3152 Pa, 3182 Pa, and 3208 Pa, respectively.

[0179] While the invention has been described in connection with what is presently considered to be the most practical embodiment, it is to be understood that the invention is not to be limited to the disclosed embodiment, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and the scope of the appended claims.

Claims

1. An extracorporeal blood circuit comprises: a blood withdrawal line connectable to an inlet of a primary chamber of a filtration unit; a blood return line connectable to an outlet of the primary chamber, the blood withdrawal line and the blood return line being configured for connection to a cardiovascular system of a patient; one or more air separators located on the blood return line and/or on the blood withdrawal line; a dialysis fluid supply line connected to an inlet of a secondary chamber of the filtration unit, a dialysis effluent line connected to an outlet of the secondary chamber of the filtration unit; one or more infusion lines connected to the blood return line and/or the blood withdrawal line; a flexible bag for enabling blood flow connected to the blood return line or to the blood withdrawal line, wherein the flexible bag comprises: an inlet port; an inlet tubing connected to the inlet port, wherein the inlet tubing is connected to the blood return line or the blood withdrawal line; an outlet port; an outlet tubing connected to the outlet port, wherein the outlet tubing is connected to the blood return line or the blood withdrawal line; a proximal end and a distal end opposite the proximal end, both the inlet port and the outlet port being arranged at the proximal end, a first film and a second film sealed to one another, and a fluid conduit, delimited by the first film, the second film and lines of welding, to put the inlet port in fluid communication with the outlet port, the fluid conduit comprising: an inlet section connected to the inlet port on an opposite side of the inlet tubing, an outlet section connected to the outlet port on an opposite side of the outlet tubing, wherein the inlet section and the outlet section are arranged at the proximal end of the flexible bag, one or more deflection sections, a connection section located upstream from at least one of the one or more deflection sections, and an additional connection section located downstream from the at least one or more deflection sections, wherein the fluid conduit has a maximum width in a direction of the fluid flow through the fluid conduit, wherein at least one of the one or more deflection sections further comprises an entry section and an exit section, each respective exit section being arranged downstream, in the direction of the fluid flow, from each respective entry section, wherein a width of the fluid conduit decreases along the direction of the fluid flow through the entry section from the maximum width to a narrower width and the width of the fluid conduit increases along the direction of the fluid flow through the exit section from the narrower width to the maximum width.

2. The extracorporeal blood circuit according to claim 1, wherein the inlet section provides the fluid conduit with a transition from a diameter of the inlet port to the maximum width of the fluid conduit at the connection section, the inlet section including an inner edge and an outer edge, the inner edge and the outer edge each forming a respective inlet angle with respect to an inlet axis of

the inlet port comprised between 5° and 30° .

3. The extracorporeal blood circuit according to claim 1, wherein the outlet section provides the fluid conduit with a transition from a diameter of the outlet port to the maximum width of the fluid conduit at the additional connection section, the outlet section including an inner edge and an outer edge, the inner edge and the outer edge each forming a respective outlet angle with respect to an outlet axis of the outlet port comprised between 25° and 60° .

4. The extracorporeal blood circuit according to claim 1, wherein the inlet section and the outlet section have different shapes.

5. The extracorporeal blood circuit according to claim 1, wherein the inlet port has an inlet axis and is in direct fluid communication with the connection section, the connection section having a main development axis, the inlet axis being parallel and laterally shifted with respect to the main development axis of the connection section.

6. The extracorporeal blood circuit according to claim 5, wherein the flexible bag comprises the connection section located directly downstream of the inlet section, a deflection section of the one or more deflection sections directly downstream of the connection section, and the additional connection section located directly downstream from the deflection section, the outlet section being located directly downstream of the additional connection section, the inlet axis is laterally shifted with respect to the main development axis away from a central development axis of the flexible bag.

7. The extracorporeal blood circuit according to claim 5, wherein the flexible bag comprises the connection section located directly downstream of the inlet section, a first deflection section directly downstream of the connection section, a first intermediate connection section located directly downstream of the first deflection section, a second deflection section located directly downstream of the first intermediate connection section, a second intermediate connection section located directly downstream of the second deflection section, a third deflection section located directly downstream of the second intermediate connection section, and the additional connection section located directly downstream from the third deflection section, the outlet section being located directly downstream of the additional connection section, the inlet axis is laterally shifted with respect to the main development axis towards a central development axis of the flexible bag.

8. The extracorporeal blood circuit according to claim 1, wherein the outlet port has an outlet axis and is in direct fluid communication with the additional connection section, the additional connection section having an additional main development axis, the outlet axis being parallel and laterally shifted with respect to the main development axis of the additional connection section.

9. The extracorporeal blood circuit according to claim 8, wherein the flexible bag comprises the connection section located directly downstream of the inlet section, one deflection section directly downstream of the connection section, and the additional connection section located directly downstream from the deflection section, the outlet section being located directly downstream of the additional connection section, the outlet axis is laterally shifted with respect to the additional main development axis away from a central development axis of the flexible bag.

10. The extracorporeal blood circuit according to claim 8, wherein the flexible bag comprises the connection section located directly downstream of the inlet section, a first deflection section directly downstream of the connection section, a first intermediate connection section located directly downstream of the first deflection section, a second deflection section located directly downstream of the first intermediate connection section, a second intermediate connection section located directly downstream of the second deflection section, a third deflection section located directly downstream of the second intermediate connection section, and the additional connection section located directly downstream from the third deflection section, the outlet section being located directly downstream of the additional connection section, the outlet axis is laterally shifted with respect to the additional main development axis towards a central development axis of the flexible bag.

- 11.** The extracorporeal blood circuit according to claim 1, wherein the inlet section has a diverging region, increasing a width of the fluid conduit from a diameter of the inlet port to the maximum width of the connection section.
- 12.** The extracorporeal blood circuit according to claim 1, wherein the outlet section has a converging region, decreasing a width of the fluid conduit from the maximum width of the additional connection section to a diameter of the outlet port.
- 13.** The extracorporeal blood circuit according to claim 1, wherein the inlet section has an asymmetrical configuration, presenting an inner edge and an outer edge, both positioned at an angle with respect to an axis of the inlet port.
- 14.** The extracorporeal blood circuit according to claim 13, wherein either or both: the angle of the inner edge and the angle of the outer edge with respect to the axis of the inlet port are different from one another, the inner edge and the outer edge of the inlet section have different lengths.
- 15.** The extracorporeal blood circuit according to claim 1, wherein the outlet section has an asymmetrical configuration, presenting an inner edge and an outer edge, both positioned at an angle with respect to an axis of the outlet port.
- 16.** The extracorporeal blood circuit according to claim 15, wherein either or both: the angle of the inner edge and the angle of the outer edge with respect to the axis of the outlet port are different from one another, the inner edge and the outer edge of the outlet section have different lengths.
- 17.** The extracorporeal blood circuit according to claim 1, wherein the inlet section has an inner edge and an outer edge and the outlet section has an inner edge and an outer edge, the inner edge and the outer edge of the inlet section forming an angle with respect to an edge of the connection section smaller than an angle formed by the inner edge and the outer edge of the outlet section with respect to an edge of the additional connection section.
- 18.** The extracorporeal blood circuit according to claim 1, wherein the inlet tubing comprises an inlet connector and the blood return line comprises an inlet counter connector placed downstream a filtration unit, the inlet connector being configured to couple with the inlet counter connector of the blood return line.
- 19.** The extracorporeal blood circuit according to claim 1, wherein the outlet tubing comprises an outlet connector and the blood return line comprises an outlet counter connector placed downstream a filtration unit, the outlet connector being configured to couple with the outlet counter connector of the blood return line.
- 20.** The extracorporeal blood circuit according to claim 1, wherein the inlet tubing comprises a Luer connector arranged at an opposite end of the inlet tubing and the outlet tubing comprises a Luer connector arranged at an opposite end of the outlet tubing.
- 21.** The extracorporeal blood circuit according to claim 1, wherein the inlet tubing comprises a male Luer connector and the outlet tubing comprises a female Luer connector.
- 22.** The extracorporeal blood circuit according to claim 1, wherein a ratio between the maximum width of the fluid conduit and a maximum height of the fluid conduit is greater than 10.
- 23.** The extracorporeal blood circuit according to claim 1, wherein, before use, the flexible bag has a flat shape and, during use, the first film and the second film cease contacting each other along the fluid conduit, thereby opening up the fluid conduit to allow fluid flow.
- 24.** The extracorporeal blood circuit according to claim 1, the fluid conduit is provided with a constant maximum width and is straight along the connection section and the additional connection section and wherein the connection section and the additional connection section are arranged parallel to one another.
- 25.** The extracorporeal blood circuit according to claim 1, wherein the inlet section comprises a first plane and a second plane, opposite to the first plane, developing from the inlet tubing towards the fluid conduit and converging towards the fluid conduit.
- 26.** The extracorporeal blood circuit according to claim 25, wherein the first plane and/or the second plane of the inlet section defines an angle with respect to a longitudinal extension of a fluid

conduit main plane less than 20°.

27. The extracorporeal blood circuit according to claim 25, wherein the inlet section further includes a terminal portion connecting the first plane and the second plane with the fluid conduit, the terminal portion including two respective layers substantially parallel at a distance corresponding to a height of the fluid conduit, the first plane and the second plane being connected to a respective layer of the terminal portion.

28. The extracorporeal blood circuit according to claim 1, wherein the first film and the second film form a portion located on the distal end of the flexible bag opposite to the inlet port and outlet port and adjacent to the fluid conduit, wherein the portion is in the form of a tab emerging from the distal end of the flexible bag opposite to the proximal end where the inlet port and outlet port are located.

29. The extracorporeal blood circuit according to claim 1, wherein the flexible bag further includes an identification region located at the proximal end of the flexible bag, being configured for receiving a readable label, and comprised at least between the inlet port, the inlet section, the outlet port and the outlet section, wherein portions of the fluid conduit are located outside the identification region.

30. The extracorporeal blood circuit according to claim 1, wherein the flexible bag, the inlet tubing and the outlet tubing are made from polyurethane or polyvinylchloride.

31. The extracorporeal blood circuit according to claim 1, further comprising: a filtration unit having a primary chamber and a secondary chamber separated by a semi-permeable membrane, the blood withdrawal line being connected to an inlet of the primary chamber and the blood return line being connected to an outlet of the primary chamber, a bypass line connecting the dialysis fluid supply line and the dialysis effluent line and bypassing the filtration unit, a dialysis fluid source comprising one or more bags of fresh fluid connected to the dialysis fluid supply line, and an infusion fluid sources comprising one or more bags of fresh fluid connected to the one or more infusion lines.

32. The extracorporeal blood circuit according to claim 1, wherein at least one of the one or more deflection sections further comprises an intermediate section interposed between the entry section and the exit section and the intermediate section has a constant width.

33. An extracorporeal blood circuit comprises: a blood withdrawal line connectable to an inlet of a primary chamber of a filtration unit; a blood return line connectable to an outlet of the primary chamber, the blood withdrawal line and the blood return line being configured for connection to a cardiovascular system of a patient; a flexible bag for enabling blood flow connected to the blood return line or to the blood withdrawal line, wherein the flexible bag comprises: an inlet port having an inlet axis, an inlet tubing connected to the inlet port, wherein the inlet tubing has an inlet tubing axis coincident with the inlet axis and is connected to the blood return line or the blood withdrawal line, an outlet port having an outlet axis, an outlet tubing connected to the outlet port, wherein the outlet tubing has an outlet tubing axis coincident with the outlet axis and is connected to the blood return line or the blood withdrawal line, a proximal end and a distal end opposite the proximal end, both the inlet port and the outlet port being arranged at the proximal end, a first film and a second film sealed to one another, and a fluid conduit, delimited by the first film, the second film and lines of welding, to put the inlet port in fluid communication with the outlet port, the fluid conduit comprising: an inlet section connected to the inlet port on an opposite side of the inlet tubing, an outlet section connected to the outlet port on an opposite side of the outlet tubing, wherein the inlet section and the outlet section are arranged at the proximal end of the flexible bag, one or more deflection sections, a connection section having a main development axis and being located upstream from at least one of the one or more deflection sections, the inlet axis being parallel to the main development axis of the connection section, and an additional connection section having a main development axis and being located downstream from the one or more deflection sections, the outlet axis being parallel to the main development axis of the additional connection section,

wherein at least one of the one or more deflection sections further comprises an entry section and an exit section, each respective exit section being arranged downstream, in a direction of the fluid flow, from each respective entry section, wherein a width of the fluid conduit decreases along the direction of the fluid flow through the entry section from a maximum width to a narrower width and the width of the fluid conduit increases along the direction of the fluid flow through the exit section from the narrower width to the maximum width.
